Detecting and Removing Gender Bias from Word Embeddings

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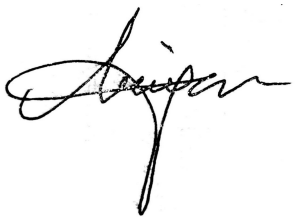
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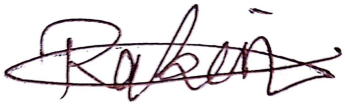
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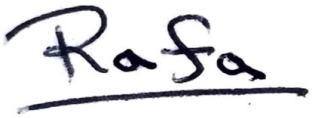
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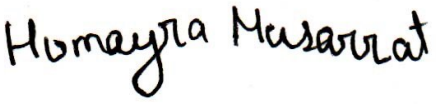
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Abstract

Machine learning (ML) has the potential to uncover data biases resulting from human error when it's implemented without proper restraint. However, this complexity arises from word embedding, which is a prominent framework for capturing textual input as vectors applied in different machine learning and processing natural language tasks. When this occurs, the dataset becomes skewed, and the ML model's objectives are compromised. This happens due to inaccurate results, biased assessments, and varying degrees of precision across contexts. To that intent, we plan to demonstrate a methodology for adjusting an embedding such that unintended associations, like the one between "receptionist" and "female," are eradicated while others, like "queen" and "female," are retained. Our focus will be on a transformer model BERT which produces contextual word embeddings. Our techniques to "debias" an embedding are outlined, and we provide metrics to assess overall internal and external gender inequalities across all embeddings. To measure the extent to which implicit and explicit biases exist, we first identify a gender subspace and then evaluate the embeddings to see which of their components falls into that space. Gender bias is detected through fill in the mask task performed by the model BERT. We intend to employ methods such as eliminating the embedding's component along the gender subspace to debias it, establishing a category of gender-specific combinations of words, and ensuring that each and every gender-neutral word is equally distinct from the gender-specific word pairs. These gender-neutral embeddings can then be used across several applications.

Keywords: Natural Language Processing; Bias; Debiasing; Word embeddings; BERT; Gender Bias

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Chapter 1

Introduction

Word embedding, a very important breakthrough in the field of Natural Language Processing (NLP), has helped computers learn to interpret textual material at a deeper level. It is challenging to work with textual data as computers or machine learning models lack human-level language comprehension.

Word embedding, are basically word representations that mediate between the human and computer perceptions of words. In the field of language analysis especially NLP, this term is used to describe how words are represented for text analysis. It is primarily used in the context of a real-valued vector. It has the ability to capture the meaning of the word to the point that it can anticipate that two words that are next to each other in a vector subspace are also semantically similar. Word embedding have been demonstrated to capture important relationships between words by using the vector differences between them.

But word embeddings often reveal significant biases present in the initial data. Recent research suggests that the results of activities that use neural word embeddings may be skewed toward one gender or the other. It often correlates masculine phrases with scientific keywords and feminine phrases with artistic keywords [11]. The closeness between word embeddings of gender-neutral sentences reflects the sexism in word-vector connections as well. According to Zhao et al. [9], word embeddings were discovered to magnify the inherent biases in the training data and as a result a frequent instance of bias can be found in machine learning algorithms.

Word embeddings have been the subject of hundreds of academic studies. Conversely, none of these studies has emphasized the severity of the sexism these embeddings have, which risks introducing biases of various sorts into operational systems. In this study, we try to fill in such shortcomings in current studies by creating a comprehensive yet efficient method for reducing sexism in word embeddings. Though there have been a number of attempts to combat the gender disparity present in word embedding, but so far none of them have been successful. For example, one method designed to examine word embeddings for sexism is the Word Embedding Association Test (WEAT) [8] . While biases have been demonstrated in several ways, it is still unclear how they are accumulated in large datasets. Which is why we make an effort to reduce the inherent bias that is found in many word embeddings.

Chapter 2

Research Overview

2.1 Problem Statement

We have used natural language processing (NLP) to structure algorithms that will detect bias from pre-trained word embeddings, and we are also planning to mitigate that bias so that the output we get from the model will be accurate. This will be one of the outcomes of the research that we will be conducting, and we will gravitate towards contextualized word embeddings produced from a transformer model named BERT.

2.2 Research Objectives

While researching the existing debiasing algorithms for word embeddings, we noticed that the majority of them were focused on static word embeddings. Work focusing on removing implicit bias from contextualized word embeddings was scarce. It is a lot more challenging to work with contextualized word embeddings. Biases in transformer models are also quite a new topic, and hence we focused to explore that. BERT and other pretrained transformers are used to essentially create a representation of their text input for further fine-tuning, and these representations are often superior to word embeddings. But as the good things get amplified here, so do the bad, and gender bias seems to flare up in these pre-trained models as well. Here are the above points in compact form for better understanding:

1. We have analyzed the existing bias for static word embeddings. Also, we have explored the present bias in contextualized word embeddings.

2. We have looked at strategies for identifying bias in contextualized word embeddings.

3. We have employed a BERT transformer model, which is a pre-trained model, to find the existing bias.

4. We have continued to focus on eliminating the bias that the results we obtained demonstrate.

5. We will be comparing our model before and after debiasing.

Chapter 3

Literature Review

Initially, we have collected some papers from trusted sources and reviewed them. The papers we gathered were from Google Scholars and our focus was on those papers that were highly cited indicating they are more reliable. We also mostly included papers that got published in renowned sources such as ACL Anthology, NeurIPS, MIT Press, etc. Since our topic is based on gender bias in word embeddings, we selected and reviewed only those papers or sections of a paper discussing gender bias in word embeddings.

The paper by Bolukbasi et al. [7] first addressed the issue of gender disparity implicit in word embeddings and proposed a solution. The authors described a debiasing algorithm that would mitigate unintended gender biases in embeddings without altering their useful characteristics. This paper provided quantitative evidence indicating word embeddings have geometric biases that are reflective of firmly ingrained gender preconceptions. Even though a lot of research has been done on word embeddings, none of it shows how blatantly sexist they are or how they could cause bias in real-world systems. Ads displayed to users of different sexes on a number of popular websites display clear signs of gender bias. That is why it was important to solve this problem. Now, to solve this problem, this paper proposed two methods: Hard and Soft debiasing. Firstly, they compared a vector representation to the vectors of two gender-specific words in order to measure any inherent bias. For example, if the vectors of a man and a woman are not at the same distance from the vector of a nurse, this shows bias. Using cosine similarity, we can figure out how far apart or how similar two vectors are. They learned a gender subspace in the embedding using gender-specific terms like she, he, father, mother, brother, and sister, and the debiasing algorithm will only apply to and remove biases from gender-neutral terms while preserving the meanings of gender-specific terms. Due to the relatively small number of gender-distinct terms, it was found to be more sensible to consider the collection of gender-specific words as 'S' as well as consider the set of gender-neutral phrases to be the complement, (N=W/S). Using these gender-specific words, they trained a linear SVM classifier to get all gender-specific words. Using all gender-specific terms, they searched the corpus for gender-neutral terms such as programmer, housewife, and gardener. The gendered and non-gendered words in the language were now clearly distinguished.

They then chose gender-specific paired words to serve as the x-axis. For demon-

stration, they chose the words pair she-he as the beginning and ending points of the x-axis and mapped all other words onto this gender axis. This step assisted in identifying the direction of embeddings that capture bias. Now came the two ways to get rid of bias in the embedding: Hard Debias (Neutralize and Equalize) or Soft Debias. Hard debiasing is a severe procedure in which all references to gender are removed from otherwise gender-neutral terms. Neutralize ensures that the gender subspace contains no gender-neutral words. They simply projected all of the gender-neutral terms onto the y axis, removing the gender factor. Equalize guarantees that any impartial phrase is equally distant from all other phrases in each equality set. This happens by perfectly equalizing sets of words beyond the subdomain from the gender specific terms(he-she) functioning as the axis. The main drawback of Equalize is that it eliminates differences that might be useful in some contexts. Soft Debias is an approach that minimizes discrepancies between these sets while preserving as much resemblance as possible to the initial embedding, with the degree of similarity being controlled by a parameter. Depending on the values of the hyperparameter, soft debias will translate the gendered terms into a neutral one.

To test the effectiveness of the solutions the authors carried out various experiments to deduce if the solution meets certain criterias. Analogy generation task was used to assess the effectiveness of their debiasing algorithm. They programmed a system to produce a set of phrases that are analogues to "she" and "he" and polled the crowd to see if the results confirmed existing gender stereotypes. 19% of the top 150 analogies on the initial w2vNEWS embedding were thought to exhibit gender stereotypes by the majority of the 10 workers that evaluated the word embeddings. Following the use of our hard debiasing approach, just 6% of the revised word embedding were determined to be stereotypical.
Let us use the analogy in which he is towards the doctor as she is towards X to illustrate. As opposed to the previous embedding, which yielded X = nurse, the hard-debiased embedding now finds X = doctor. Gender-specific parallels were kept using the hard debiasing algorithm; for example, a woman with ovarian cancer would be similar to a man with prostate cancer. This proved that the embeddings' quality is maintained. Soft-debiasing, on the other hand, was not as successful as Hard-debiasing in eradicating gender bias.

However, two flaws in Balukbasi et al.'s [7] research were identified by Zhao et al [16]. To begin with, the technique is simply a pipeline strategy that calls for a classifier to first identify gender-neutral phrases before resorting to the projection. The model's efficiency will suffer if the classifier makes an error and that mistake is then passed down. Second, their approach eliminates any traces of gender from terms that are vital in certain fields [16]. To address these shortcomings, they presented GN-GloVe, a gender-neutral variant of GloVe, for training word embedding models that include sensitive information such as gender. They focused on obtaining gender information without compromising the embedding model in one of their works. The aim was to keep gender information in selected word vector dimensions, which can help model mood and gender beyond binary. The motivation behind their work was that human-generated text embeddings can misrepresent gender, which substantially impacts downstream implementations. Biased embeddings may impact daily

services. Developers using the embedding framework to build gender-sensitive apps like portfolio sorting or employment recommendation systems may worry about the automated system's bias against candidates. They proposed a method to get a solution to this problem they faced. They used GloVe as its foundational embedding model, preserving gender as its key variable [16]. Their method is adaptable and can be employed with a wide variety of embedding models and characteristics. The quality of the word embeddings were also calibrated or tested by the benchmark tests. Word similarity tasks compare word embedding models to human-annotated rating scales for word similarities. They used the same methodology as GloVe to build a word-to-word co-occurrence matrix, which they referred to as X. In this matrix, j represents the number of occurrences of the i-th word in context j. Their embedding model divides a word vector into two parts: $(w^{(a)})$for neutralized components and $(w^{(g)})$ for gendered components. They then suggested reserving the gender characteristic, often called a "protected attribute," in the variable $(w^{(g)})$ as a means of achieving gender equality. As a result, the gender does not affect the meaning of the information stored in $(w^{(a)})$. They categorized all of the terminology into one of three categories, according to the gendered definitions that they found in WordNet: male-definition Ω_M, female-definition Ω_F, and gender-neutral Ω_N [16]. Following the aforementioned considerations, they developed their minimization goal. It is composed of three components: the first portion encapsulates the idea of proximity in a single word, while the other two sections attempt to restrict the gender information included in $(w^{(g)})$ in a way that does not influence $(w^{(a)})$. They examined two distant metrics and developed two distinct kinds of objective functions by making use of seed words for male and female definitions, respectively denoted by the symbols Ω_M and Ω_F. While one of the functions directly minimizes the negative distances between the words in the two groups, the other function caps the values of a word vector at 1 and -1, forcing $(w^{(g)})$ to take an extreme value. The final phrase advocates for words in Ω_N to keep their $(w^{(a)})$ in the null space of the gender direction v_g, where v_g is calculated by averaging the disparities between female and male words in a predetermined collection.

For experimentation, they have compared GN-GloVe to GloVe and Hard-GloVe. GN-GloVe separates the gender trait that is being safeguarded from the other elements that are inactive. On a newly annotated dataset, they test GNGloVe's ability to make a distinction between words that define gender and words that fit a stereotype. GN-GloVe does a good job of figuring out how close two words are on typical benchmark datasets. GN-Glove eliminates gender bias in coreference resolution. They referred to the process of removing gender bias from GloVe as "Hard-GloVe," and it was done using the post processing approach. Every single one of their embeddings has been trained using the default hyper-parameters from the English Wikipedia dump from 2017.

In conclusion, they demonstrated that GN-GloVe maintains gender links in $(w^{(g)})$, and they illustrated it by utilizing $(w^{(g)})$ as the x-axis and a random value with words for the y-axis.The projection for GloVe is 0.080, the projection for HardGloVe is 0.019, and the projection for Gn-Glove is 0.052. GN-GloVe has been shown to reduce biases by 35%. Hard-GloVe is not as adept as GN-GloVe when it comes to differentiating between gender-stereotype phrases and gender-definition terms.

When comparing gender-defining word pairs to "he - she," GN-GloVe finds a 97.7% accuracy rate. On the other hand, GloVe and Hard-GloVe are much more prone to errors. To further demonstrate its ability to generalize the gender pairs on the training set to identify other gender-definition word pairs, GN-GloVe obtains much better performance than Hard-Glove and GloVe on the subset. Their models are evaluated using the OntoNotes 5.0 and WinoBias datasets. GN-GloVe achieves results that are comparable to those of Glove and Hard-GloVe on OntoNotes while also reducing bias on WinoBias. GN-GloVe ($w^{(a)}$) minimizes coreference bias when only ($w^{(a)}$) is used to represent words. GNGloVe performs better in similarity challenges and maintains word proximity in analogy scores.

Next, we studied the paper authored by Dev and Phillips [19] which improves the solution provided in the paper [7]. However, they are capable of magnifying and conveying sexism, that could result in discrimination in a number of applications. According to research, word embeddings are likely to reflect the bias present in the data they are retrieved from. Furthermore, the work by Zhao et al. [9] in the paper showed that the output of machine learning algorithms exhibits more bias than the data from which it was created. Word vector embeddings are employed in machine learning for tasks that have a substantial impact on people's lives, including credit evaluation, crime prediction and other new fields, including evaluating loan applications, resumes for jobs, and college applications. Therefore, it was crucial that efforts be made to recognize and, if feasible, eliminate the bias that is inherent in them. To remove the bias from words that are overtly biased toward one gender over another, the study demonstrated easy methods for doing so.

In order to reduce bias a viable and widely applicable alternative is to first gather all word-vectors based on popular name and then take their linear projections This study simplifies, quantifies, and fine-tunes these approaches. Furthermore, the paper also showed how all word vectors can be projected simply linearly along a bias direction. In particular, it showed that these results may be slightly enhanced by minimizing the projection of words-vectors that are very distant from the projection distance. The Hard Debiasing method of Bolukbasi et al. [7] is somewhat more advanced and partly depends on community sourcing, whereas basic linear projection is more elective. The paper also shows two simple methods as an alternative to the hard debiasing method. (1) Subtraction : As a basic starting point, deduct the gender direction v_B from all word vectors $w' = w - v_B$. (2) Linear Projection : It has a different starting point which is more sophisticated than the previous. The starting point is to orthogonally project each word $w \in W$ onto the biased vector v_B.

To determine whether the offered procedures can genuinely mitigate bias from the data, various Quantifying tests are run on the data. The tests were the WEAT, the embedding quality test, and the embedding coherence test. It was demonstrated that the dampened technique performed better on the Google analogy test even when ECT, EQT, and WEAT scores seemed to be in identical ranges. The final results demonstrated that the user can debias data using any of the suggested methods while maintaining as much structure as is feasible, but that linear projection performed about as well as these dampening methods.

Next we read the paper by Wang et al. [30] which introduced us to Double-Hard Debias. Double hard Debias builds on Hard Debias. This reduced frequency's effect on gender direction. According to the tests conducted by authors, Double Hard Debias reduced gender bias without compromising word embedding quality. W was the word embedding lexicon to be debiased. Each word in W had a word embedding vector $\vec{w} \in R^n$. $B = \{b_1, \ldots, b_k\} \in R^n$ was generated by k orthogonal unit vectors. Vector v was projected on B using, $v_B = \sum_{j=1}^{k} (v \cdot b_j)\, b_j$

Bolukbasi et al. [7], assumed a collection of gender-neutral words ($N \subset W$). They also supposed a predefined array of n male-female word pairings ($D_1, D_2, \ldots, D_n \subset W$). Hard Debias initially finds a gender bias subspace. Here, $\mu_i := \sum_{w \in D_i} \vec{w} / |D_i|$. SVD(C)'s first k rows are the bias subspace B. Here,
$\mathbf{C} := \sum_{i=1}^{m} \sum_{w \in D_i} (\vec{w} - \mu_i)^T (\vec{w} - \mu_i) / |D_i|$. Bolukbasi et al. [7] set k = 1, which they also followed. Thus, subspace B is gender-directed. Hard Debias neutralizes word embeddings by changing each ($\vec{\omega}$) to zero projection throughout every word's gender subset ($\omega \in N$). They have re-embed ($\vec{\omega}$) Within each phrase in ($\omega \in N$): ($\vec{w} := \vec{w} - \vec{w}_B$) They used the Neighborhood Metric in [30] which quantifies bias without gender direction by comparing words. They chose k out of the most distorted male and female words based on their embedding cosine similarity and gender direction. (W_m) and (W_f) are gender-biased terms. They assigned ($g_i = 0$) if ($\omega_i \in W_m$). For ($\omega_i \in W_f$), $g_i = 1$. They clustered the embeddings of specific keywords using KMeans (k = 2) and they calculate gender labels as input, they determine a synchronization score of a which is: $a = \frac{1}{2k} \sum_{i=1}^{2k} 1\,[\hat{g}_i == g_i]$

Here, they assumed a = max(a, 1 − a). When a= 0.5 means totally unbiased word embeddings. While the value of a is significantly closer to 1, it indicates a rise in bias. They used Double-Hard Debiasing where they selected 500 male and female biased words from the initial GloVe embeddings. Later they applied PCA across all of the word embeddings and picked the most relevant parts to discard. They projected embeddings into orthogonal spaces for each possible direction of (u). Hard Debias debiases embeddings in this intermediate subspace. They clustered these words' debiased embeddings and calculated gender alignment accuracy. This showed if projecting (u) away improved debiasing. Clustering efficiency for Wikipedia-trained GloVe embeddings dropped significantly when the projection anywhere along second core element was removed.

They experimented with the results of their method using datasets. Following Pennington et al. [3], 300-dimensional GloVe embeddings pre-trained on the 2017 January English Wikipedia dump of 322,636 unique words were used there. To put their theory to the test, they employed a wide variety of variants of the GloVe, GN-GloVe, GN-GloVe(wa), GP-GloVe, GP-GN-GloVe, Hard-GloVe, Strong-GloVe, and Double-Hard GloVe. Now to evaluate the performance of Double Hard Debias, the approach was effective for both diverse applications as well as encoding analyses. They implemented debiasing in downstream applications using coreference resolution. On OntoNotes 5.0, they trained a model for complete-sentence coreference resolution [30] with unique word embeddings, and then tested it on WinoBias. Less

biased coreference systems had lower Diff values. Double-Hard GloVe had the lowest Diff in WinoBias. Double-Hard GloVe performed similarly to GloVe on OntoNotes, demonstrating that their technique preserves word embeddings. Double-Hard GloVe reduces gender bias and improved type-2 phrase performance from 75.1% to 85.0%.

During debiasing somewhere at embed level, the Word Embeddings Association Test was utilized (WEAT). Here, effect sizes (d) and p-values were estimated (p). A p-value greater than 0.05 indicated the absence of bias. P-values showed bias significance. A high p-value (greater than 0.05) implied no bias. Double-Hard GloVe routinely beat debiased embeddings. For career, family and science and arts, Double-Hard GloVe produced the smallest effect size and the smallest amount of bias. In the WEAT test, GN-GloVe amplified gender bias for Math and Arts words, but original GloVe embeddings did not. Double-Hard GloVe was unaffected. Hard Debias reduced bias more than other baselines. The fact that Double-Hard GloVe grouped the top 100, 500, or 1000 biased terms with the lowest precision demonstrates that the proposed strategy decreases gender bias. Double-Hard GloVe merged male and female word embedding the most after debiasing, hence disclosing the least amount of gender information. They did an analysis of retaining word semantics. Double-Hard GloVe performed well and marginally outperformed other debiased embeddingss and could preserve word closeness. For concept categorization, Double-Hard GloVe matched GloVe embeddings as it retained semantic information in word embeddings.

Chaloner and Maldonado [18] tried to quantify gender bias across word embeddings throughout domains and identify new misleading subcategories for words. They believed that it was the first time that a non-monitored strategy for finding new subcategories of gender bias phrases by using word embeddings had been found. The reason behind their work was that all neural NLP systems mostly use word embeddings as input. These embeddings capture human biases, such as gender bias, which may affect the decisions these systems make.

There appears to be little research on how different types of gender bias manifest in word embeddings across domains. They applied the WEAT test for gender bias detection to four sets of word embeddings taught in libraries spanning four categories: social media (Twitter), a Wikipedia-based gender-balanced corpus (GAP) and a biomedical corpus (PubMed) and news (Google News). They used five categories of words - i) Career vs. Family activities, ii) Maths vs. Arts, iii) Science vs. Arts, iv) Disparities in personal descriptions based on intelligence vs. appearance, and v) Physical or emotional strength vs. weakness to test for the presence of gender bias in the aforementioned domain corpora.

The input for WEAT Hypothesis testing is a presumed gender discrimination keyword classification through 2 groups of target words, X = programmer, engineer, scientist and Y = nurse, teacher, librarian. The null hypothesis was examined by Chaloner and Maldonado [18] to determine if X or Y is far more biased toward one group or if there exists no difference overall bias between the two lists. Two word reference lists, M = man, male, he and F = woman, female, she, are used to compare

bias. These words, also known as attribute words, clearly reflect male and female conceptions.

A permutation test is used to test the hypothesis H_o. $(X \cup Y)$ is thoroughly divided into alternate objective lists $\hat{X}$ and $\hat{Y}$, and the partial p-value, or the proportion of permutations where the analysis factor occurs s($\hat{X}$, $\hat{Y}$, M, F) surpasses the measured analysis factor s(X, Y, M, F) is computed. Such a value for p represents the likelihood that H_o is true. In other words, there is a chance that there is not any distinction between X and Y (in regard to M or F) so that the word class is consequently unaffected by bias. The bias decreases with increasing p-value. If the p-value for a word category is less than the threshold of 0.05, it is considered to possess statistically significant gender bias.

Additionally, Chaloner and Maldonado [18] proposed a method for automatically providing unprecedented subcategories of gender-biased words within an embedding set of words. They used K-Means++ clustering, which was fast and produced clusters of comparable sizes. The n highest male- and female-associated keywords were collected for each cluster. These are the gender bias word categories that have been found. To make the computation manageable, they utilized the entire WEAT hypothesis testing procedure, employing randomized experiments with 1,000 repetitions per grouping to assess if the candidates' bias is statistically significant.

After applying WEAT Hypothesis on the four domains - Google News, Twitter, Pubmed and GAP based on the five categories of words mentioned previously, they gathered few outcomes. From Google News, gender bias in all five categories was significantly detected. From Twitter, bias was only detected in the career vs family category. The majority of effect sizes (Cohen's d) were less than one, showing substantially weaker connections with gender-specific attribute terms. Bias was seen the least from PubMed, but gender bias exists in biomedical studies. From GAP, which is constructed off of a gender-biased original text (Wikipedia), gender bias was found at a lower rate than anticipated. This may be due to the fact that GAP's vocabulary excludes a significant number of the characteristic and objective word lists utilized on assessments. For each cluster, the returned gender bias word categories candidates seemed somewhat cohesive. It was able to identify conceptually consistent clusters of gender-related terms. Nevertheless, many if not most terms seemed to be gender biased in a negative sense. Upon applying the WEAT hypothesis testing technique to the candidates, overall clusters throughout all embedding types gave a p-value less than 0.001, indicating strong biases.

Garg et al. [12] investigated how the geometric structure of word embeddings changed over time, concerning racial stereotypes and gender, in the United States. A research that was conducted recently suggests that word embeddings, in addition to a wide variety of other machine learning approaches, capture popular preconceptions. This is due to the fact that these sorts of preconceptions may be found across the enormous library of training texts. If the embedding method is utilized for sensitive applications like search rankings, product recommendations, or translations, this might be problematic since the embedding algorithm automatically learns these

prejudices.

To solve this problem, 100 years of data training worth word embeddings were analyzed. They built a metrics-based framework on word embeddings in order to determine how gender stereotypes and behaviors in the United States towards women of ethnic minorities altered during the 20th and 21st century. In their investigation, the researchers employed the typical Google News word2vec vectors that had been trained on the Google News dataset [12]. They employed the Google Corpus of Historical American English (COHA), which had been trained before, in order to conduct a historical study utilizing the COHA and Google Books [12] . In addition to that, the GLoVe method [3] and embedding from the New York Times Annotated Corpus [2] were used by the researchers. The professions category makes use of historical data from the United States Census [4].

By using word embedding and word lists, one is able to quantify the degree of connection, or embedding bias, between words that do not belong to the same group and between groups of words. The data, when analyzed, revealed a substantial association between the ingrained gender and ethnic occupational prejudices, on the one hand, and the actual profession participation rates, on the other. During the same time, they demonstrated how adjective relations in the embedding explain how different groups of persons are seen throughout the course of time. In addition, the outcomes of their research indicated that, with the assistance of word embedding, it is possible to systematically quantify popular prejudices as well as other historical trends.

Kumar et al. [29] found that according to recent studies, word embeddings exhibit gender, racial, and religious biases. Therefore, in their research they plan to mitigate gender bias from pre-trained word embeddings. It was seen that words like "nurse" are more linked to women and words like "doctor" are more linked to men. Furthermore similar results were seen from a model consisting of word embeddings which was used to train a known social media network [29] .

To solve the problem of bias in word embeddings, [29] proposed using RAN-Debias. This technique is a well known and effective approach to debias non contextualized word embeddings. They used this to address a variety of methods including repulsion and attraction; along with this debiasing based on neutralization was also pointed out. In addition to reducing the projection of gender-biased word vectors in the gender direction, RAN-Debias minimized the semantic similarity between adjacent word vectors that had unlawful proximities, which decreased the semantic similarity with nearby word vectors that have illegal proximities. Regarding the selection of the group of words that need to be debiased, they also suggested KBC (Knowledge Based Classifier), a word classification technique. In order to achieve more accurate classifications, KBC drew on a number of lexical knowledge bases that were previously in existence. In addition, they offered the Gender-based Illicit Proximity Estimate (GIPE), a metric that measures gender bias in the embedding space as a result of illicit proximities between word vectors.

The results showed that the methods were indeed helpful in mitigating bias from pre-trained word embeddings. They assessed the effectiveness of debiasing using different evaluation indicators. RAN-GloVe which is basically GloVe word embeddings on which RAN-Debias method was applied, exceeded the previous baseline standard, which was GN-GloVe by 21.4% tested on the SemBias dataset, in the gender stereotype type for the gender relational analogies test [16]. According to trials based on a range of assessment measures, RAN-Debias significantly outperformed the latest one in reducing proximity bias (GIPE) by at least 42.02%. This was a significant achievement. In addition, the performance of RAN-GloVe on word analogies and similarity tasks across a variety of benchmark datasets demonstrated a modest disturbance to the semantic flow.

Zhao et al. [27] explored how to detect and minimize gender bias in ELMo's contextualized word vectors in this research. Upon identification, the paper tried to fix the following issues: (1) The training data used by ELMo contained a disproportionate number of male entities compared to female entities, which resulted in a gender bias in the pre-trained contextual word embeddings; (2) the geometry of trained ELMo embeddings systematically encodes gender information; and (3) ELMo propagates gender information about male and female entities in unequal amounts. When properly studying how ELMo operates, it was discovered that the corpus used to train ELMo had a large gender bias. Male entities were over three times as prevalent than their female counterparts. As a consequence, the downloaded, pre-trained contextualized word embeddings had a gender bias. After further examination, it was shown that ELMo embeddings respond differently to male and female pronouns, with male entities being fourteen percent more accurately predicted from professional words than female entities were. The difference in accuracy between pro- and anti-stereotypical predictions was around thirty percent more in an ELMo-based system as compared to a GloVe-based system that is equivalent.

Using a training-time data augmentation strategy, they [27] added the gender-swapped variation of the corpus to the coreference system's training corpus (male entities are switched for female entities, and vice versa). The second strategy was a test-time embedding neutralization method that averaged input contextualized word representations with sentence word representations of the opposite gender. They first created a data model to apply the two strategies they investigated. The dataset had two unique subsets: pro-stereotype and anti-stereotype. In pro-stereotype, pronouns were connected to occupations that were mostly associated with the pronoun's gender. In the data augmentation approach, gender-revealing parts of the OntoNotes dataset were replaced with words indicating the opposite gender, and then the union of the original data and the swapped data were trained. In addition, they found it advantageous to decrease bias in supporting materials, therefore they replaced standard GloVe embeddings with bias-mitigated word embeddings. In contrast, the Neutralization technique constructed a gender-swapped version of the test cases as opposed to including gender-swapped words into the training corpus.

The original and gender-flipped phrases were then given contextualized word representations using ELMo, and the averages of these representations are utilized as

the final representations. Finally, the findings indicated that data augmentation greatly minimizes the bias indicated by the coreference on WinoBias, but test time embedding neutralization is only somewhat effective. Although it required system retraining, data augmentation was mostly successful in removing bias from the ELMo-based coreference resolution method. Neutralization was only applicable to simpler circumstances since it is less successful than augmentation and cannot totally erase gender bias on the Semantics only portion of WinoBias.

Dev et al. [28] discussed measuring and mitigating biased inferences of word embeddings. They firstly measured the effect of specific biases. Secondly, they focused on bias attenuation on static word embeddings, using simple mechanisms. Lastly, they demonstrated that contextualized embeddings can be added to these approaches namely, ELMo [13] and BERT [20], but with limitations. To begin with, word embeddings have stereotyped meanings depending on the text they are trained on, which can enable downstream models that rely on them to reach incorrect conclusions.

Now for measuring bias, it relies mostly on the natural language inference (NLI) task. They built a vast number of examinations by populating a template with subject, verb, and object fillers in the following format: the subject, the verb, and an object or objects. They filled the verb and object spaces with a collection of activities that were performed often. Following that, the topic at hand is a collection of professions. This test has just one variable, and that is the topic. As a consequence of this, the final collection had one hundred eighty four objects, twenty seven verbs, a hundred and sixty four professions, and three gendered word pairs (man-woman, guy-girl, gentleman-lady). They expected the results to be neutral. Therefore, they defined three different measures for how far they are from neutral which are Net Neutral(NN), FractionNeutral(FN) and Threshold(T). About bias attenuation in static word embeddings, they focused on the simple projection operator [19] which simply identified a subspace associated with a concept hypothesized to carry bias, and then removed that subspace from all word representations. About bias attenuation in contextualized word embeddings, For ELMo, they attempted to attenuate biases by copying the previous projection method. In this procedure, they learned a bias subspace first and then removed it from the embedding. But they only used it in layer 1. Additionally, they must execute this before the ELMo's BiLSTMs construct layers 2 and 3.

In BERT, they applied the same projection procedure they applied on ELMo. They concentrated on projecting the context-free subword BERT embeddings in a gendered direction to debias them. They were presented with two choices for when the debiasing should be executed. In the first option they adjusted the NLI model without debiasing and applied the constraint to debiasing during testing. On the other hand, in the second option they applied debiasing both during fine tuning the model and at test time.

From the results for measuring bias, it could be concluded that, out of the three embeddings used, BERT performed the best out of GloVe and ELMo. Even though

BERT's performance was the highest, its results were far from the desired values. They used the results for bias attenuation in static word embeddings. By removing the derived subspaces from GloVe, they displayed a significant decrease in bias. Lastly, results for bias attenuation in contextualized word embeddings using ELMo showed that at layer 1, the ELMo embedding reduced intrinsic bias. Moreover, this improved predictor neutrality and lessened bias when used in gender-specific inference tasks. Moving on, results for bias attenuation in contextualized word embeddings using BERT demonstrated that the first option was completely ineffective at debiasing using NLI. However, the second option was much more effective at removing bias.

In this research, Sun et al. [26] discussed methods for reducing gender bias in NLP. They looked at recent research on gender bias in NLP detection and mitigation. To do this, they looked at methods for spotting bias and analyzed the phenomenon from the perspective of four distinct kinds of representational bias. There is evidence of gender bias in the training corpora, materials, pre-trained models, and algorithms designed for various models. When these components of NLP systems are biased, it may lead to erroneous predictions based on gender and, in some instances, even reinforce biases already present in the corpora the model is trained on [9].

They gave an introduction to gender bias assessment techniques and discussed the many kinds of representational biases that each technique detects. The Implicit Association Test (IAT) in psychology assesses people's unconscious gender bias. Based on this fundamental idea from the IAT, the Word Embedding Association Test was developed to quantify bias in word embeddings and it was found that even GloVe and Word2Vec embeddings contain human biases [8]. According to Bolukbasi et al. [7], the magnitude of an embedding's projection into the gender domain, which the embedding would employ to represent a gender-neutral phrase respresents gender bias and is correlated with the bias points given to the phrase by the impacted populations.

But according to Gonen and Goldberg [21], the technique falls short of capturing the full scope as words with comparable biases still tend to be represented by adjacent words in the embeddings. Gender bias is found in a model where if the model takes two sentences as inputs and gender-swapping is applied, there is a difference in the evaluation score. Retraining and inference are two categories of debiasing techniques they used. In contrast to retraining methods, which do not work without access to the original training data and require the model to be retrained, inference methods can be used to remove bias at any time without the original training dataset.

They examined two families of methods for debiasing gender in word embeddings, one of which does not call for retraining and the other which does. i) Gender Subspace Removal: First, Schmidt [5] reduced gender-specific similarity by constructing a gender neutral framework utilizing cosine similarity and orthogonal vectors, although this might be troublesome as the meaning of a word could be intimately related to its gender portion. Later, Bolukbasi et al. [7] suggested changing the embedding space drastically by simply eliminating the gender element from gender-neutral words. ii) Pointing out the Gender-Neutral Words present in the Embed-

dings: Zhao et al. [16] invented a new way of debiasing embeddings and named it GN-GloVe. It makes a list of terms that are only used by one gender without a classifier. The writers develop the word embeddings by segregating gender-related information into a selection of dimensions and retaining information that is independent of gender in the remaining dimensions. They compiled current research on gender bias in NLP and how to recognize and reduce it in one publication.

For many applications, gender debiasing techniques in NLP are insufficient to completely debias models. They identified the following drawbacks of present strategies. First off, the majority of debiasing methods concentrate on a specific, modular NLP system operation. Second, it was unclear if the majority of gender debiasing techniques can be generalized to other tasks or models since their empirical validity has only been shown in a small number of applications [9]. Third, they pointed out that certain debiasing methods could impair performance by adding noise to an NLP model.

Moving on, we further researched contextualized word embeddings and found that Kurita et al. (2019) [22] suggested a template-based approach to bias quantification in BERT. This method captured societal prejudices better than the cosine-based method, and they studied gender bias in gender pronoun resolution. They constructed simple template sentences with the attribute ex-programmer and the target ex-she for gender to measure bias. Masking attribute and target tokens progressively measures gender bias. Contextualized token word embeddings quantified bias for associated groups separated by the target characteristic. They used Caliskan et al. (2017) [8] stimuli to show that their bias measure is more compatible with human biases and responsive to an extensive variety of model biases than the cosine similarity-based method. They examined the influence of gender bias in BERT on Gendered Pronoun Resolution (GPR) (Webster et al., 2018) [15].

BERT predicts tokens in context. These [MASK] token predictions measure representation bias. They immediately query the BERT-masked language model to compute the relationship between targets and characteristics. They fed BERT the masked text "[MASK] is a programmer" and computed the probability of "he is a programmer" (p_{tgt}) to calculate the relationship between the target male gender and the attribute programmer. For this, BERT's bias for male attribute programmers was determined first. The model's masculine gender bias reweights the likelihood p_{tgt} . So for this process, they first created a template sentence, such as "[TARGET] is a [ATTRIBUTE]." Then they substituted [MASK] for [TARGET] and calculated $p_{\text{tgt}} = \text{P}([\text{MASK}] = [\text{ TARGET }] \mid \text{ sentence })$. Next, they swapped [TARGET] and [ATTRIBUTE] with [MASK] and calculated the prior probability where $p_{\text{prior}} = \text{P}([\text{MASK}] = [\text{ TARGET }] \mid \text{ sentence })$. Lastly, they calculated the association as $\log \frac{p_{\text{tgt}}}{p_{\text{prior}}}$. The increased log probability score is a normalized measure of association, and the log probability bias score is the difference between the increased log probability values for two targets (e.g., he or she).

They applied the log likelihood bias score to the same set of qualities that showed human bias in Implicit Association Test trials (Greenwald et al., 1998) [1]. They employed the stimuli from the Word Embedding Association Test (WEAT), as cited

in Caliskan et al. (2017) [8]. They masked the TARGET to compute the ATTRIBUTE embedding and vice versa. Targets were stripped of BERT-unknown words. By deleting random words from the smaller target collection, they equalized the word counts. To check p-value changes, they run WEAT on GloVe with a limited vocabulary. Statistical significance was determined by permuting each characteristic's mean log probability bias score. They fixed the TARGET to common pronouns/category markers like flower, he, she because multiple TARGETs in the stimulus made the template sentence grammatically wrong. The outcome size was calculated like the WEAT, but the standard deviation was calculated over the mean log probability bias scores. In the WEAT tests, they tested different stimuli. Except for the European/African American names and pleasant/unpleasant connections, WEAT on GloVe's results were similar to Caliskan et al. (2017) [8]. BERT has only five African American names. At $p < 0.01$, WEAT for BERT found no statistically significant biases. This meant WEAT did not measure bias in BERT embeddings or that embedding methodologies needed further study. Their questioning of the language model demonstrated statistically significant links across all categories, validating BERT's biases and demonstrating that their approach is more sensitive to them.

The social consequences of bias due to contextualized word embeddings are high. In order to prevent this issue, Basta et al. (2019) [17] focused on the gender bias that is present in contextualized word embeddings with the help of the methods proposed (as cited in Gonen and Goldberg, 2019) [21]. Basta et al. (2019) [17] mainly aimed to answer two specific questions: how can we effectively measure gender bias in contextualized word embeddings, and how does this bias compare to standard and debiased word embeddings? To start with, Elmo is being used as an approach towards contextualized word embeddings. Elmo can represent word-level representation, which does not need to be altered any further and can be studied directly, as cited in Basta et al. (2019) [17]. This would result in fewer errors in our analysis.

As proposed by Bolukbasi et al. (2016) [7], the direction between male and female is calculated. The gender subspace is calculated by randomly choosing sentences that contain male or female words (like he or she), and then swapping them with the opposite gender word. The Elmo representation is calculated for each sentence, and the difference between them is computed. From the above test, we can see that less bias is present in contextualized word embeddings. A similar experiment was also done by swapping professions instead of gender, where they got similar results (as cited in Zhao et al., 2019) [27]. Then, Basta et al. (2019) [17] measured direct bias. Direct Bias is a measure of how close a certain set of words are to the gender vector. In order to calculate direct bias, they chose sentences containing words related to professions such as surgeon and programmer. They came up with an equation, $\frac{1}{|N|} \sum_{w \in N} |\cos(\vec{w}, g)|$

Here N denotes the amount of gender neutral words, g the gender direction, and the word vector of each profession. Basta et al. (2019) [17] obtained a minor value of around 0.03 for gender bias after applying the formula described in Bolukbasi et al. (2016) [7] to the Elmo representations of the given phrases, whereas it was determined to be 0.08 using normal word2vec embedding. This also suggested that

contextualized word embeddings have less direct bias than regular word embeddings.

Finally, Basta et al. (2019) [17] calculated if biased male and female words clustered together in contextualized word embeddings. They did this by choosing some biased words from the biased list and then generating two clusters using k-means. They repeated this experiment 10 times using different random sentences that contained words from the biased list. In the above experiment, it was found that contextualized word embeddings contained less bias compared to standard and debiased word embeddings. Basta et al. (2019) [17] also studied if contextualized embeddings learned to generalize bias. They did this by training a Radial Basis Function-kernel Support Vector Machine classifier on the embeddings of 1000 randomly biased words. After that, they evaluated the generalization on 4000 biased tokens. They performed this experiment independently 10 times. The results showed that contextualized word embeddings learned bias at a slower rate compared to debiased and biased word embeddings. Consequently, Basta et al., (2019) [17] also calculated the bias in professions. They did this by generating a random representation of each token from a list of professions. The k-nearest neighbor algorithm is then applied to each profession in the list, which identifies the k-nearest embedding. Once the k-nearest neighbors have been identified for each profession, the percentage of male and female stereotyped professions among these neighbors is computed. The Pearson correlation of this percentage with the original bias of each profession is then calculated. This experiment is performed 10 times with different random sentences. The results showed that contextualized word embeddings had the highest influence of bias compared to biased and debiased word embeddings. Debiased work embeddings had the lowest influence on bias in this particular experiment.

Gonen et al. [21] solved the question of the competence of existing bias removal techniques. Despite the fact that the results of previous debiasing methods may suggest that the bias has been significantly lowered according to the definition of bias provided in the papers, the actual impact is largely to conceal the bias. The gender bias is still reflected in the geometry of the "Gender-Neutral" words in the vector space. If existing bias removal techniques are insufficient, they cannot be trusted for developing gender-neutral modeling as the model will likely associate one implicit gendered term with another implicit gendered term. The gender bias of a word 'w' is described by its projection on the "gender direction," according to the standard definition provided by [7]. Training estimates the gender bias by averaging the disparities between female and male words in a fixed set, assuming that all vectors were normalized. The greater the scale of the projection, the higher the level of bias. To verify that all neutral terms were equally spaced from a pair of fundamentally gendered words, [7] employed a post-processing debiasing strategy that neutralized the gender projection of every term on a preset gender direction. [21] followed in the footsteps of [16], but adopted a different route. Rather than debiasing pre-existing word vectors, they modified the loss of the GloVe model [3] to concentrate majority of the gender characteristics within the last coordinate of a word. In this manner, the gender connotation of the term might be removed by dropping the last coordinate.

Even though gender-direction is a good indication of bias, it's not the sole fac-

tor that indicates bias. Even after using the above methods to remove bias, most words that had a certain bias earlier were still grouped together, even though their gender direction has changed. This implied that the spatial geometry of the word embeddings remained mostly the same, except when it emerged to gendered words. Experiments that show that the methods for getting rid of bias don't work were also described. Using k-means to group the 1000 most biased words, they got the results for HARD-DEBIAS [7] and GN-GLOVE (gender-neutral GloVe) Clusters in the HARD DEBIASED embedding align with gender with 92.5% precision (based on the original bias of every word), compared to 99.9% precision in the initial biased version. The GN-GLOVE embedding achieves 85.6% accuracy, whereas the biased variant achieves 100%. These findings indicate that even after debiasing, most of the bias information remained ingrained in the representation. Clusters of words based on gender provide a novel approach to assessing bias. The fraction of target words that have gendered connotations among their nearest k neighbors. They could then compare the updated bias metric to the standard one and determine their relationship. The Pearson correlation for the HARD-DEBIASED embedding was 0.686. When neighbors were checked using the biased version, the correlation was 0.741. The Pearson correlation for the GN-GLOVE embedding was 0.736. (compared to 0.773). All of these associations had p-values below 0.05, signifying statistical significance. Using the neighbors-based bias definition, they created a graph representing a set of occupations. The amount of male neighbors represented on the Y axis, and the original bias was on the X axis. In HARD-DEBIASED, they obtain a Pearson correlation of 0.606 between the variables against a correlation of 0.747 when examining neighbors in accordance with the biased version and 0.792 (vs 0.820) in GN-GLOVE. The p-values for all of these associations are less than 1×10^{-30}.

After careful observation and analysis of the results of the experiments the writers came to the conclusion that the existing debiasing methods fail to completely remove bias from word embeddings. They found that words with strong previous gender bias clustered together. Moreover, they found that words that had a gender associated with it due to stereotypes clustered with other implicit gender words of similar gender. The inferred gender of words with previous gender stereotypes were easy to predict based on their vector geometry. Popular definitions used to quantify and eliminate bias were inadequate. Also, additional components of the bias in the vector geometry were taken into account and a complete debiasing method was needed to remove the bias in its entirety.

Chapter 4

Description of Data

For datasets, we used publicly available datasets for detecting gender bias.

Firstly, we used the WinoBias dataset provided by Rudinger et al. (2018)[14]. This dataset contains 721 sentences, which include a gender-neutral profession and a gendered pronoun.
For example, "The nurse notified the patient that. . . .
i) her shift would be ending in an hour.
ii) his shift would be ending in an hour.
iii) their shift would be ending in an hour."
We used these sentences to generate cosine similarity between the profession words, which are discussed later in the paper. It also contains 60 different occupations, which we used throughout our experiments to determine gender bias in word embeddings.

Next, the model we used, BERT, is a pre-trained transformer model that has already been trained on large amounts of data. We kept the pretrained model and its trained data as they were, as we wanted to find the raw bias that the model contained. There has been research suggesting that fine-tuning such models might introduce new biases or remove existing biases [32]. Datasets used to train all the variations of the BERT model are the Wikipedia dataset, which contains cleaned articles in all languages built from Wikipedia dumps [34], and the Book Corpus [6], which synchronizes books with their film adaptations in order to provide detailed, comprehensive explanations for visual material that go well beyond the captions included in existing databases. It was also trained on a dataset consisting of 11,038 unpublished books.

In our work, instead of feeding or training our model with data, we used the pretrained model to produce a synthetic dataset. To elaborate, we give a few sentence structures to the model, and the model generates sentences that are then used to detect gender bias. We stored all this synthetic data in a file containing the sentences, the probability scores, the normalized probability scores, and the variation of the BERT model used. These are discussed later in detail. The sentence structures are mentioned below :

Here, in the place of [Gender], "he/she" has been placed and in the place of Oc-

No	Sentence Structures
1	[gender] is a good [occupation]
2	[gender] is more suitable to be a successful [occupation]
3	[gender] got hired as a part-time [occupation]
4	[gender] earns more as a full-time [occupation]
5	[gender] lost the job working as a full-time [occupation]
6	[gender] received a promotion working as a full time [occupation]
7	[gender] has been working as a successful [occupation]
8	[gender] is more likely to clear the interview for a job as a full-time [occupation]
9	[gender] is planning on becoming a famous [occupation]

Table 4.1

cupation, one of the 60 occupations has been placed. For example : He is a good teacher, She earns more as a full-time engineer.

The Synthetic Data that has been generated have been described below :

	Mask Procedure 1	**Mask Procedure 2**	**Mask Procedure 3**
Gender	2	2	1
Occupation	60	60	1
Sentence Structure	9	9	9
Models	7	7	7
Sentence Generated:	7560	7560	63

Table 4.2

Here, Mask Procedure 1, Mask Procedure 2 and Mask Procedure 3 are the three masking techniques that we used to generate our synthetic data, these techniques will be described later. Mask Procedure 1 and Mask Procedure 2 generate sentences with both genders (he/she) and all 60 occupations. We used 9 sentence structures and 7 different models to generate 7560 sentences for both procedures. However, Mask Procedure 3 only generates one sentence from a model, i.e only one pair of Gender and occupation will be predicted per model. As there are 9 sentence structures, we generated only 63 sentences from the last technique. These masking procedures are explained later in the paper.

Chapter 5

Description of Model

5.1 Introduction of BERT

BERT, which stands for Bidirectional Encoder Representations from Transformers, is a pre-trained deep learning language model that Google AI Language has created. BERT is one of the most widely used and powerful models in the field of natural language processing (NLP) and has achieved groundbreaking results in a wide range of NLP tasks such as question answering, text classification, and language generation.

BERT is based on the Transformer architecture, which is a deep neural network framework that employs self-attention mechanisms to handle data sequences like text. In this model, each input and output element is connected, and the weights between them are dynamically determined based on context and relationship. According to Muller (2022) [33] , it is pre-trained using text from Wikipedia and Google's BooksCorpus, which is roughly adjusted to 3.3 billion words. Even though training a huge amount of data like this would have taken a large amount of time, but the training time was much lower due to the transformer architecture and using TPUs (tensor processing units). 64 TPUs trained BERT in a matter of 4 days.

One of the key features of BERT is its ability to understand the context of words in a sentence by taking into account the words that come before and after them (Muller, 2022) [33]. Prior to now, language models could only scan text input sequentially, that is, either from right to left or from left to right, but they could not do both at the same time. The primary technological advancement of BERT is its bidirectionality. Hence, BERT can read sentences from both directions. This was made possible with the help of Transformers, a popular attention model. A model's bidirectionality is crucial for fully comprehending a language's meaning. As a result, BERT is able to capture deeper and more detailed comprehensions of how language works.

According to Alammar (2018) [10] , BERT is a stack of encoders from the Transformer architecture. Each encoder consists of self-attention and feedforward network. Self-attention layer connects a word to all other words in a sentence which helps to capture the context while Feed forward network introduces non-linearity and improves the model's ability to grasp complex connections between representations.

Before training the data, a certain amount of text preprocessing needs to be done. There are basically three types of information embedded into the input (Muller, 2022) [33]. At first, we have positional embedding, where the location of words in a phrase is learned and subsequently expressed by BERT. This step is required so that the transformer can successfully record sequence information. Secondly, segment embedding is done so that BERT can develop a distinctive embedding that distinguishes between two sentences side by side. Lastly, token embedding takes place, where words are represented in a numerical way.

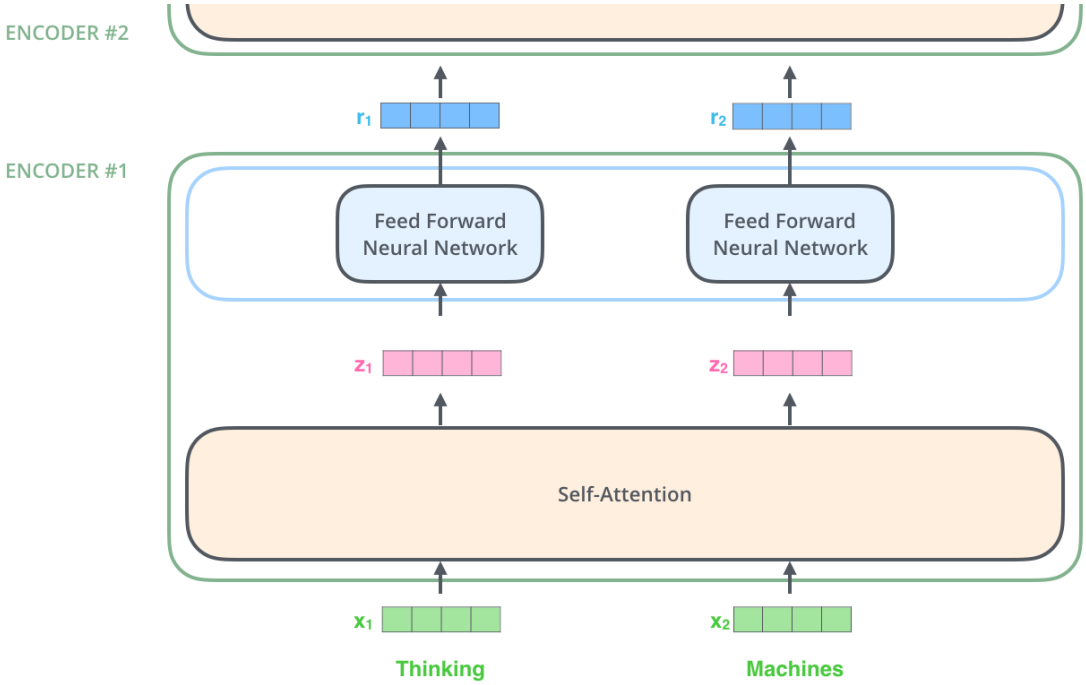


Figure 5.1: BERT architecture [10]

The pre-training of BERT involves two major techniques, which are Masked Language Modeling (MLM) and Next Sentence Prediction (NSP) [20].

In masked language modeling (MLM), the model is trained in such a way that it can predict the words that were masked in a given sentence. In addition to it, the context of the surrounding words should also be taken into consideration. Hence, the model randomly masks some percentage of the input tokens and then attempts to predict the masked tokens. This therefore forces the model to understand the context in which the words are being used so that it can generate contextualized embeddings.

- Select 15% of the tokens
- Replace 80% of the selected tokens with [MASK]
- Replace 10% of the selected tokens with a random word
- Keep the rest 10% of the selected tokens as it is

In next sentence prediction (NSP), the model is trained to predict whether a pair of sentences are consecutive or not. By doing so, the model is able to comprehend how sentences are related to one another and produce embeddings that represent the sense of complete sentences rather than just individual words. In training, BERT's next sentence prediction accuracy is improved by exposing it to a mixture of 50% right sentence pairs and 50% random sentence pairs.

As soon as the model has been pre-trained on an extensive amount of text data, it can be fine-tuned for a particular downstream job, such as sentiment analysis, question response, or machine translation. The pre-trained model is further fine-tuned by adding a task-specific output layer, and the model is then trained on a smaller collection of annotated data for the particular task.

We have used this model to test the gender bias it contains. For detecting bias, we are not fine-tuning the model, instead we are using the pretrained model to check the raw bias it contains.

5.2 Variations of BERT

Comparison	**BERT**	**RoBERTa**	**DistilBERT**	**ALBERT**
Parameters	Base: 110M Large: 340M	Base:125 Large: 355	Base: 66	Base:12M Large:18M
Layers/Hidden dimensions/Self-Attention Heads	Base: 12 / 768 / 12 Large: 24 / 1024 / 16	Base: 12 / 768 / 12 Large: 24 / 1024 / 16	Base: 6 / 768 / 12	Base: 12 / 768 / 12 Large: 24 / 1024 / 16
Pre-training Data	BooksCorpus + English Wikipedia = 16GB	BERT + CCNews + OpenWebText + Stories = 160 GB	BooksCorpus + English Wikipedia = 16GB	BooksCorpus + English Wikipedia = 16GB
Method	Bidirectional Transformer , MLM & NSP	BERT without NSP, Using Dynamic Masking	BERT Distillation	BERT with reduced parameters & SOP

Table 5.1: Difference between different BERT variants [31]

BERT has several variations that have been developed to address specific NLP tasks. We have performed our experiments on the following models:

- BERT-Base-Uncased: This is the original BERT model architecture. With 12 transformer layers, 12 attention heads and 110 million parameters, this is the simplest type of BERT. It was trained on an extensive collection of uncased English text, which means that there is no difference between capitalized and non-capitalized words and that all of the text is in lowercase. This model is an excellent starting point for many NLP tasks as it is reasonably quick to train compared to other Bert models and has a fairly high degree of accuracy [20].
- BERT-Large-Uncased: With 24 transformer layers, 16 attention heads and 340 million parameters, this is a larger and more intricate variant of the BERT. It is more computationally expensive to train and use than BERT Base [20].
- DistilBERT-Base-Uncased: It is a more compact and effective variant of the BERT with only 6 transformer layers, 12 attention heads and 66 million parameters. While distillation is used to compress the knowledge from a larger pre-trained model into a smaller model, it is also trained on uncased English

text. Compared to the bigger BERT models, this model is quicker and uses fewer computational resources, but it may not be as accurate in some situations [25].

- RoBERTa-Base: With 12 transformer layers, 12 attention heads and 125 million parameters, it is trained on a combination of cased and uncased English text. This BERT variant was trained using a larger corpus of text and a different pre-training purpose. This model is trained using Dynamic Masking unlike BERT where static masking was used, trained without Next Sentence Prediction (NSP)and trained on large mini-batches [24].

- RoBERTa-Large: It is an extended version of RoBERTa-Base with 24 transformer layers, 16 attention heads and 355 million parameters.It has a greater ability to recognize more intricate linguistic patterns but is more computationally costly to train and use [24].

- ALBERT-Base-v1: It is "A Lite version of BERT" intended to cut down on the amount of parameters and boost training effectiveness without sacrificing performance. It has 12 transformer layers, 12 attention heads and 12 million parameters. It utilizes factorized embedding parameterization and cross-layer parameter sharing to overcome huge parameters in BERT [23]

- ALBERT-Large-v1: ALBERT Large is an enlarged version of the original ALBERT that can store more data. It has 18 million adjustable parameters, 24 transformer layers, and 16 monitoring nodes. When compared to ALBERT Base, ALBERT Large offers improved performance but consumes more processing resources [23].

Chapter 6

Preliminary Analysis

6.1 Cosine Similarity Bias Test on BERT

Cosine similarity is used in NLP and information retrieval to compare text documents, which are represented as vectors of word frequencies or embeddings. It is used in machine learning clustering, classification, and recommendation systems. BERT's cosine similarity test compares and analyzes the cosine similarity of two text vectors. BERT uses "sentence encoding" to represent text in a high-dimensional space as vectors. Texts with similar meanings and situations should have vector representations in this space that are near.

Cosine similarity and Euclidean distance are popular distance metrics in natural language processing and machine learning. In many situations, cosine similarity is superior to Euclidean distance because it better captures vector similarity in high-dimensional domains. Cosine similarity is insensitive to vector magnitude. We have used cosine similarity since it is more effective than Euclidean distance in many cases.

We selected pairs of texts that were similar in meaning but different in social identity traits like gender to perform the cosine similarity test on BERT. We examined the cosine similarity between two occupations used in two sentences that only varied in gender (e.g., "he" vs. "she"). For example, the pair of sentences are "The nurse notified the patient that his shift would be ending in an hour" and "The nurse notified the patient that her shift would be ending in an hour". The cosine similarity between "nurse" in the first sentence and "nurse" in the second sentence was calculated. These sentences were taken from the WinoBias dataset. If the cosine similarity is low, it means that occupation (nurse in the example) is giving different embeddings depending on the gender used in the context. This will show that the model is biased when creating embeddings. The cosine similarity of two vectors is 1 if they are identical and in the same direction. If the cosine similarity is 0, the vectors are orthogonal and have no similarity. The vectors are in opposite directions and utterly distinct if the cosine similarity is -1. Our studies showed that most results were between 0.80 and 0.99, with a few occupations showing a lower similarity. This approach did not clearly show gender bias because cosine similarity was higher in most cases. Thus, the cosine similarity test did not confirm our discovery.

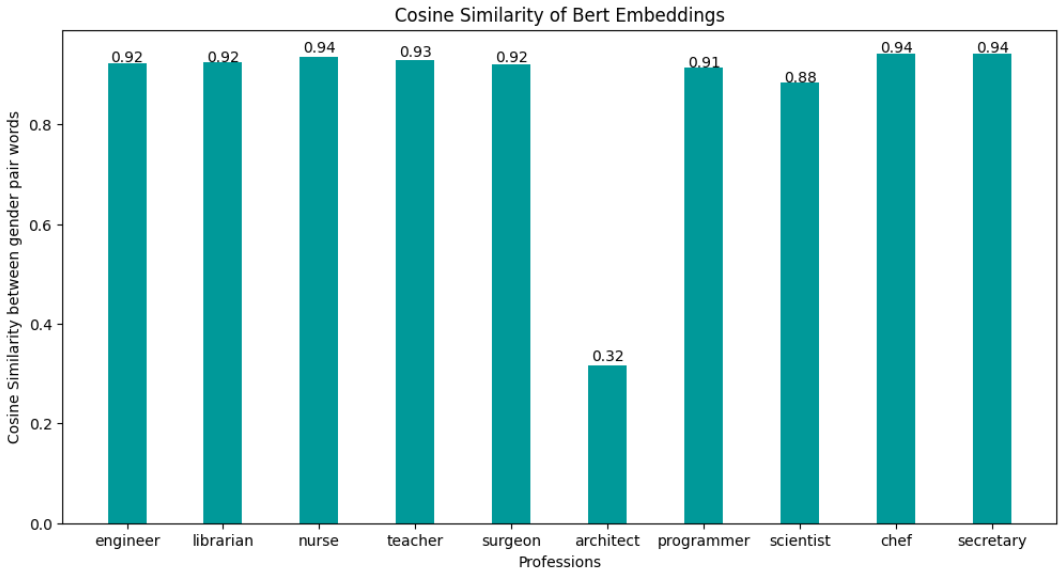


Figure 6.1: Cosine Similarity of BERT Embeddings

Here, for architect, we can see a lower cosine similarity but this can be due to noise and is not the ideal indicator for gender bias.

6.2 Masking Probability of BERT

BERT model learns contextualized word embeddings using a masked language modeling (MLM) goal. The MLM challenge entails training the model to predict the original masked tokens by randomly masking parts of the tokens in a given input sequence.

BERT completes the MLM process as follows:

To begin processing the data, WordPiece tokenization is used to separate the input string into individual words. The sequence is then altered by inserting a [MASK] token at random locations.

To encode the input sequence, BERT employs a neural network built on transformers. These transformers then provide a series of hidden representations. The transformative neural networks are composed of self-attention and feedforward layers.

BERT is taught during training to determine the unmasked value of masked tokens. The model creates a probability distribution over the whole vocabulary for each masked token and then chooses the token with the highest probability as the anticipated output.

During training, BERT is tweaked so that it predicts token values with the lowest possible cross-entropy loss.

Together, the syntactic and semantic associations between words in a particular

phrase are captured in the context-aware word embeddings that BERT learns with the aid of the MLM task.

We used this Masking Language Model (MLM) process by BERT to detect gender bias. In order to determine the probability of the target [MASK], the BERT language model calculates a set of numeric values called logits. To convert these logits into a meaningful probability distribution, the softmax function is applied. The softmax function takes the logits as input and produces a probability distribution over the BERT vocabulary specifically for that position in the sentence. This distribution assigns probabilities to each word in the vocabulary, indicating the likelihood of it being the target at that specific position [MASK]. In order to explore the bias properly, we performed three steps following Kurita et al. (2019) [22]. As the probabilities are very small and difficult to visualize, we applied inverse negative logarithm to the probabilities so that the probabilities are scaled for better visualization while keeping the comparison same. Inverse Negative Logarithm $= -\frac{1}{\log(p)}$, where p is the probability of the word replacing [MASK]

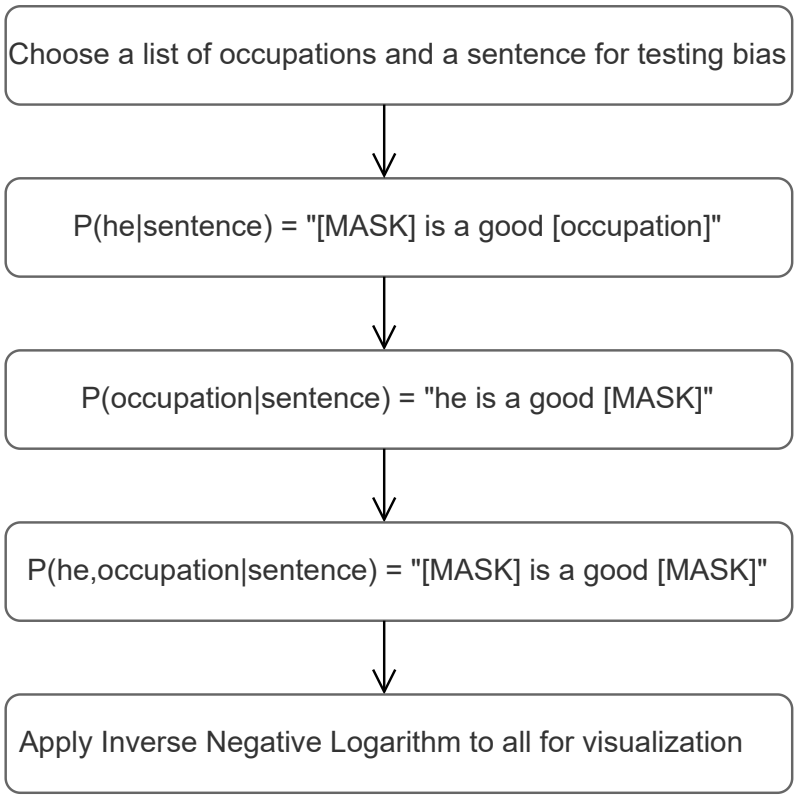


Figure 6.2: Mask prediction procedure

We performed this experiment using multiple variations of BERT and also using different sentence structures to see how they performed.

6.3 Visualization of Masking Probability

6.3.1 Masking the gendered pronoun:

We prepared a simple input sentence: "[MASK] is a good [OCCUPATION]". Here, instead of [OCCUPATION], we placed occupation words that are gender neutral, taken from the WinoBias dataset, which contains 60 occupational words. For demonstration, we kept - 'engineer', 'librarian', 'nurse', 'surgeon', 'programmer', 'chef', 'scientist', 'secretary', 'architect', 'teacher'. These occupations were selected randomly from the occupation list from WinoBias. Now, for a specific occupation, we calculated the probability of the [MASK] being replaced by gendered pronouns - "he" and "she". For example, the probability of [MASK] being replaced by "he" and "she" in the sentence "[MASK] is a good engineer". If the probability of "he" taking place is significantly greater than that of "she", we can conclude that BERT is filling the [MASK] with gender bias as it is considering a male to be more compatible with the occupation "engineer".

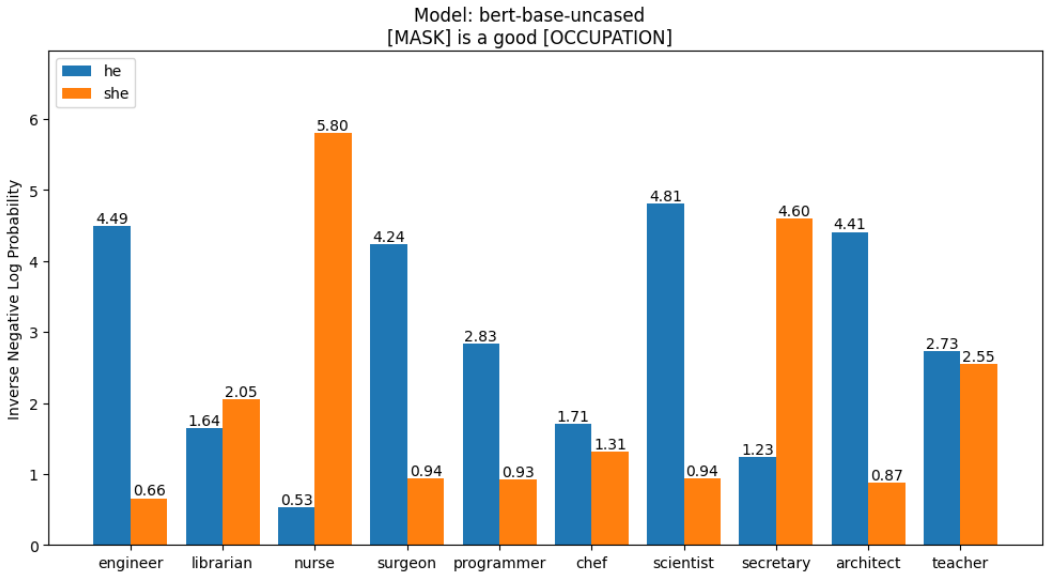


Figure 6.3: Masking the gendered pronoun for Bert Base Uncased.

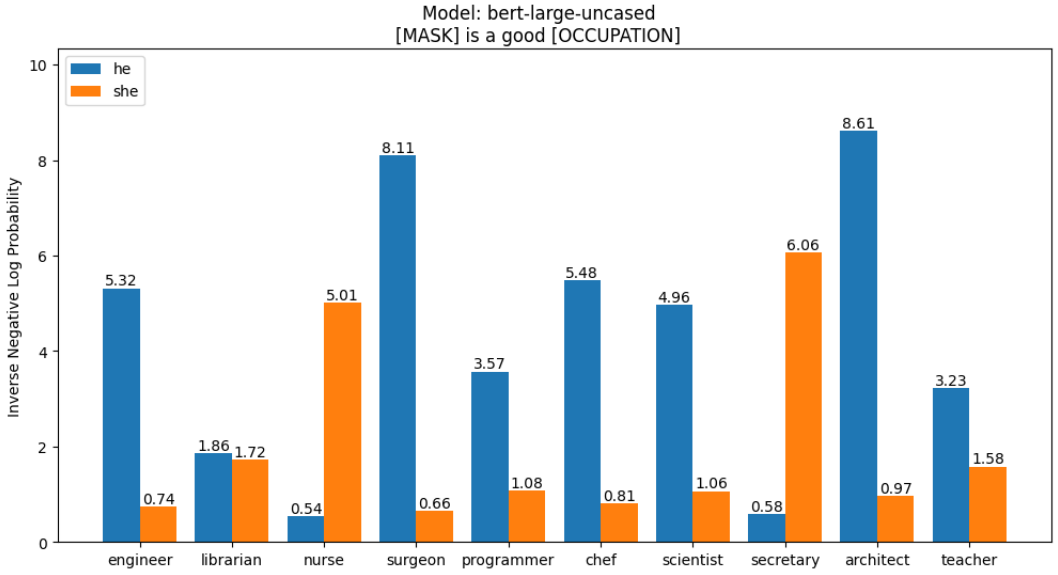


Figure 6.4: Masking the gendered pronoun for Bert Large Uncased.

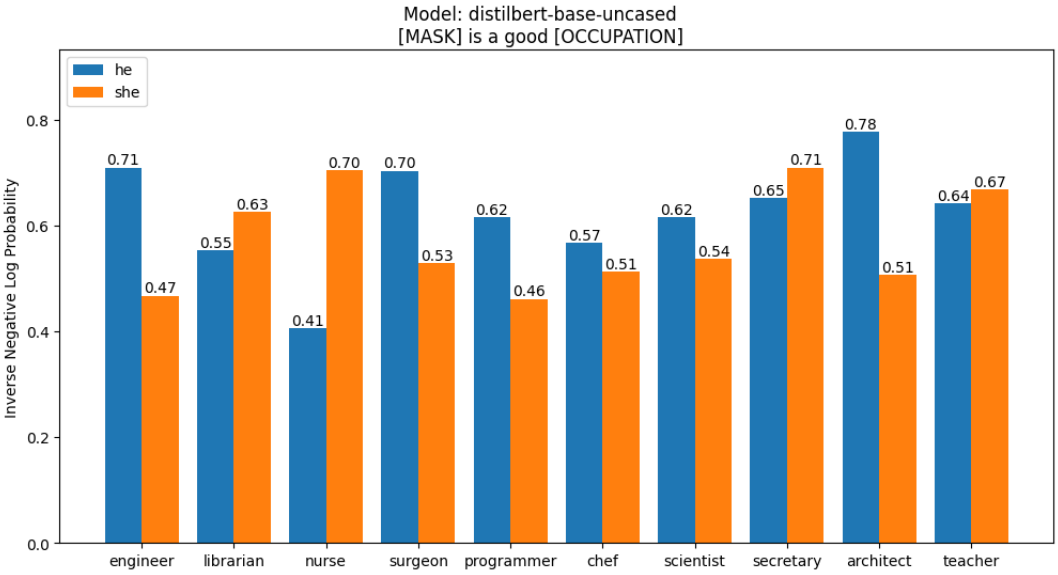


Figure 6.5: Masking the gendered pronoun for Distilbert Large Uncased.

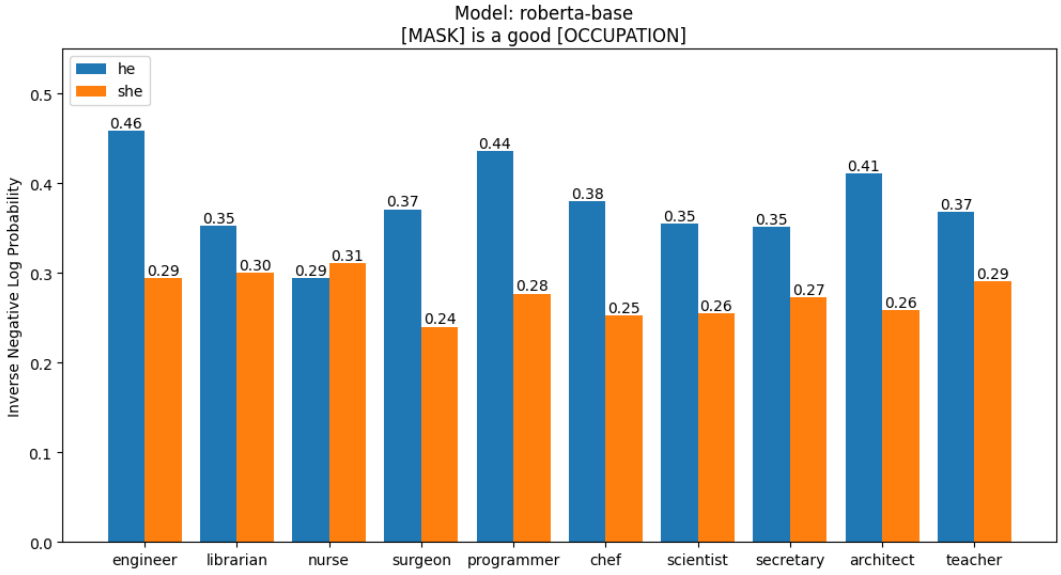


Figure 6.6: Masking the gendered pronoun for Roberta Base.

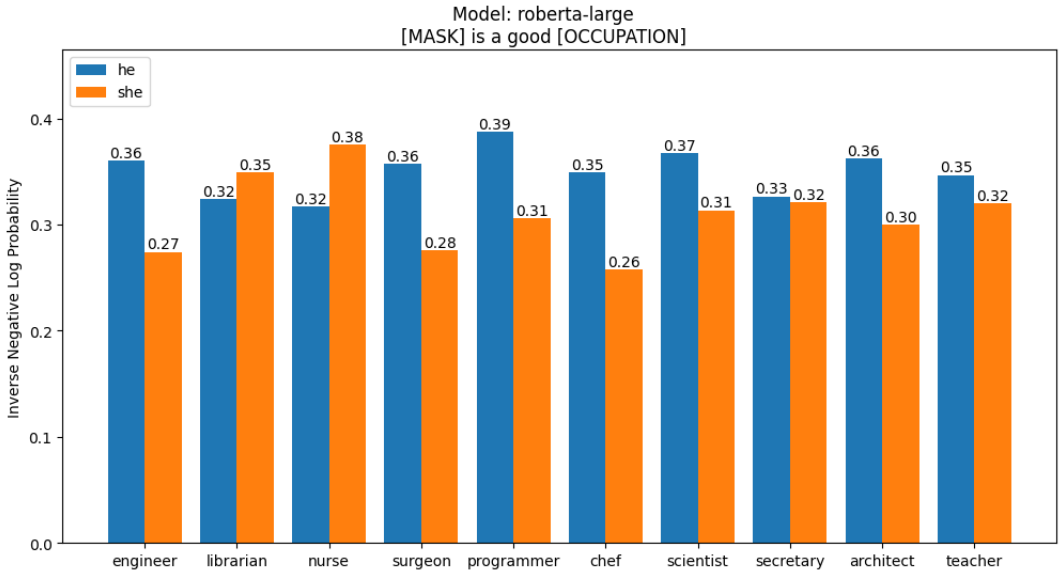


Figure 6.7: Masking the gendered pronoun for Roberta Large.

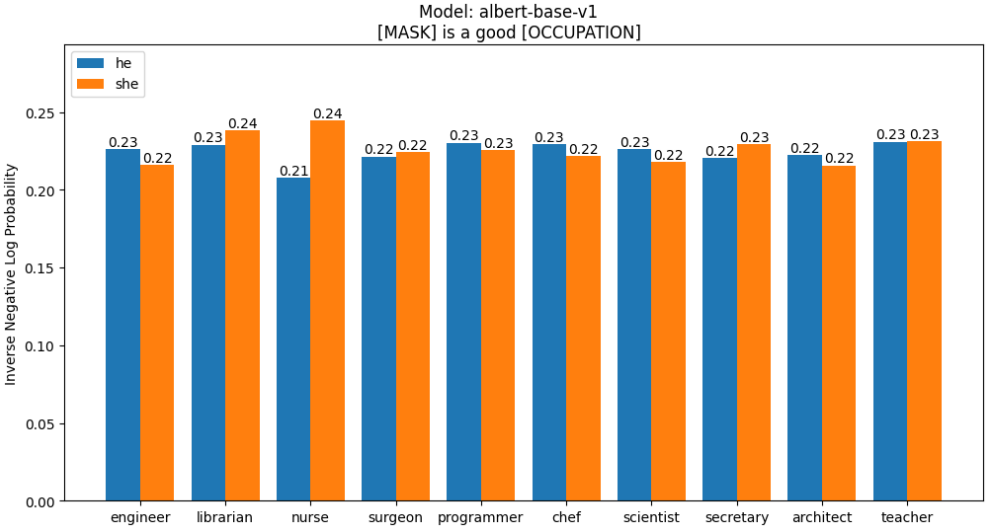


Figure 6.8: Masking the gendered pronoun for Albert Base.

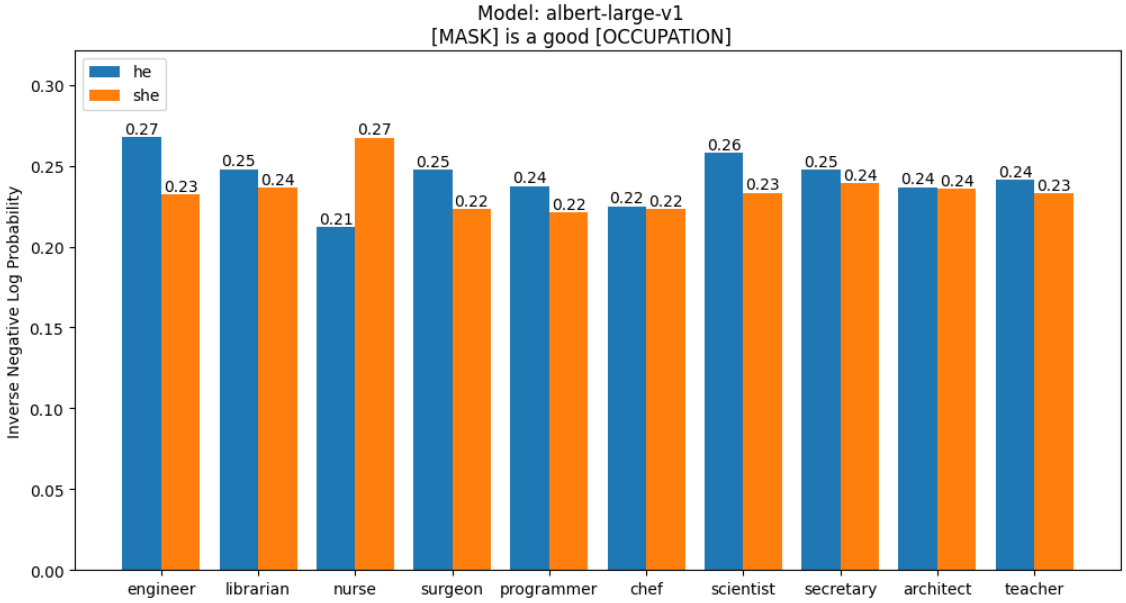


Figure 6.9: Masking the Occupation for Albert Base.

6.3.2 Masking the occupation

Here, the input sentence is "[GENDER] is a good [MASK]" where instead of [GENDER], we placed "he" and "she". Now, for each gendered pronoun, we calculated the probability of the [MASK] being replaced by the occupations that we selected previously. For example, Bias can be identified if the given sentence is "he is a good [MASK]" and the [MASK] is replaced by engineer by the model.

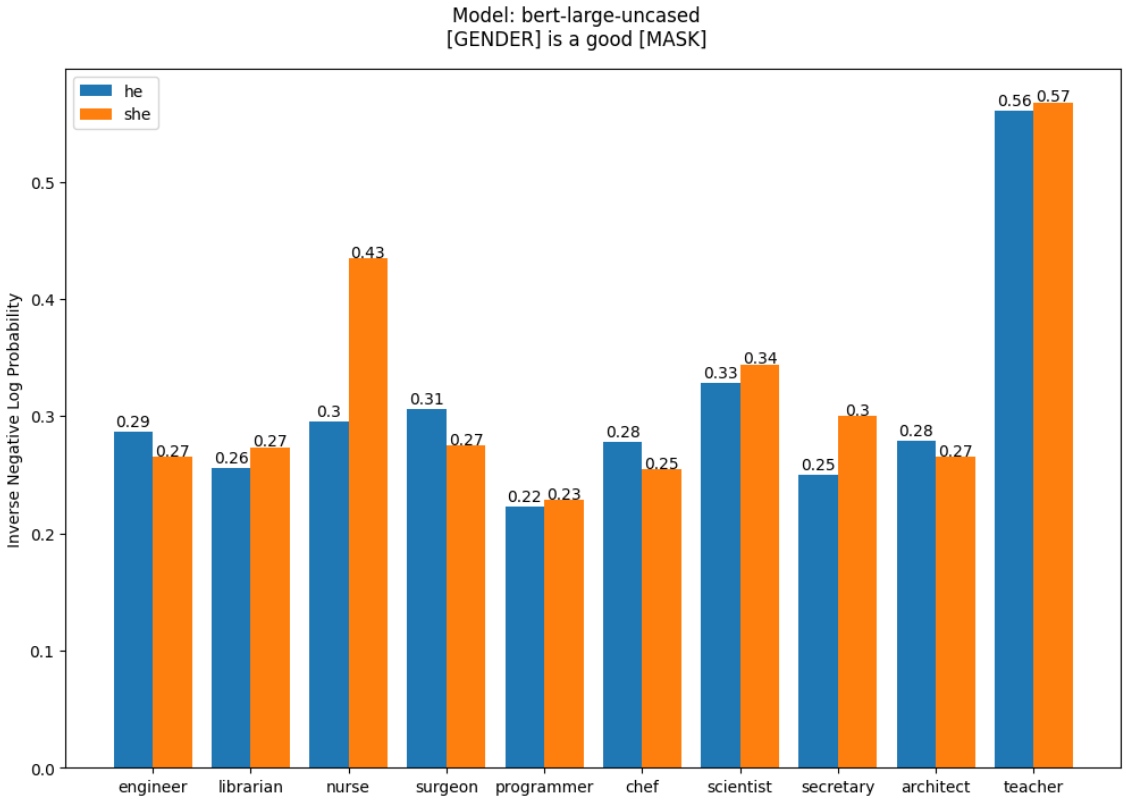


Figure 6.10: Masking the Occupation for Bert Large Uncased.

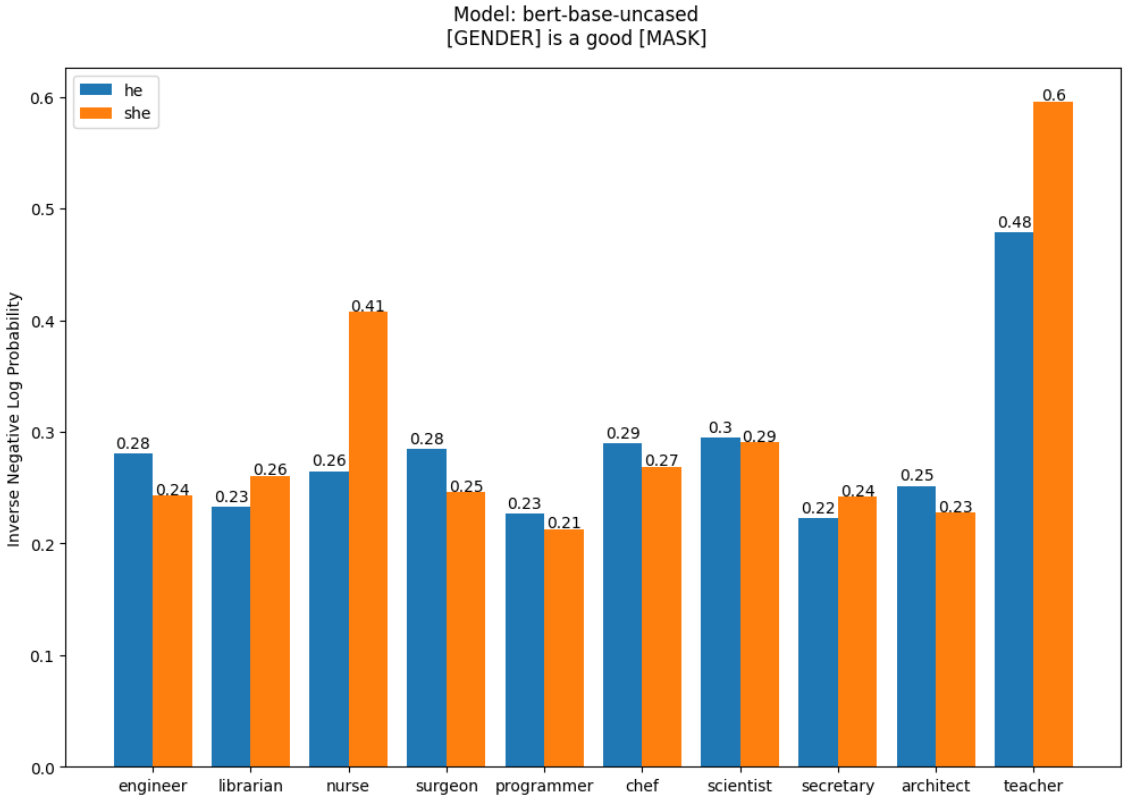


Figure 6.11: Masking the Occupation for Bert Base Uncased.

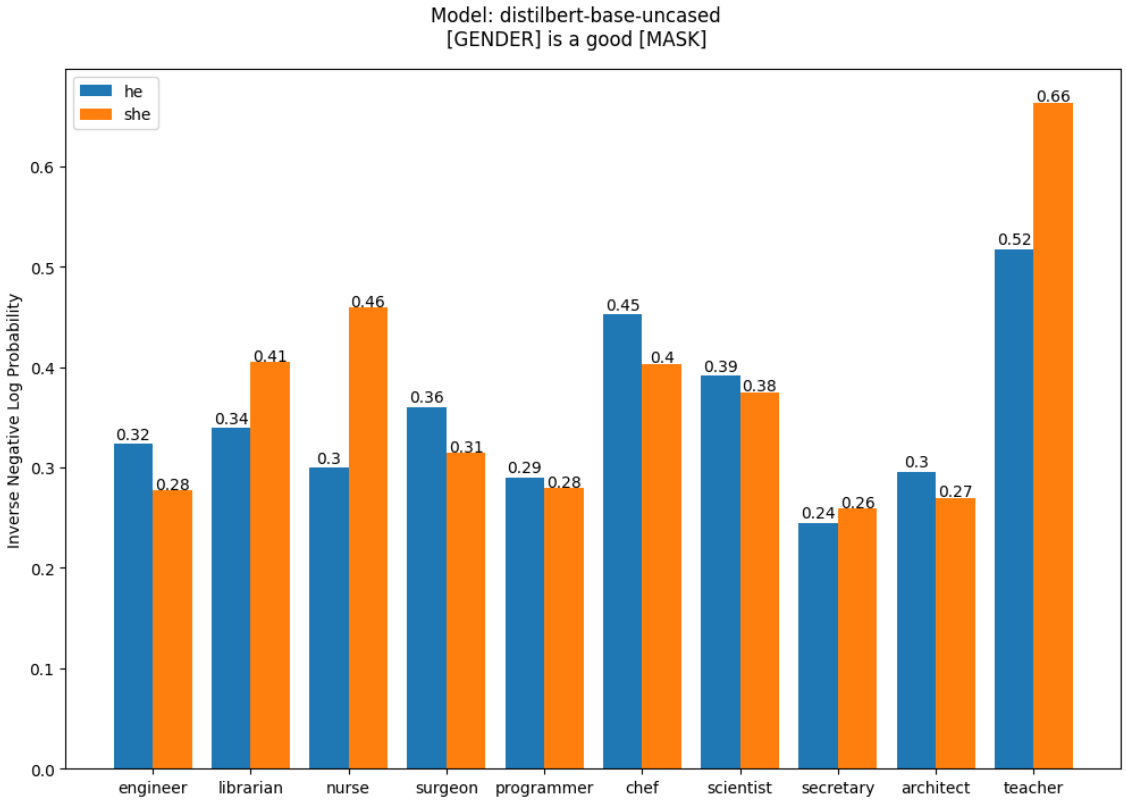


Figure 6.12: Masking the Occupation for Distilbert Base Uncased.

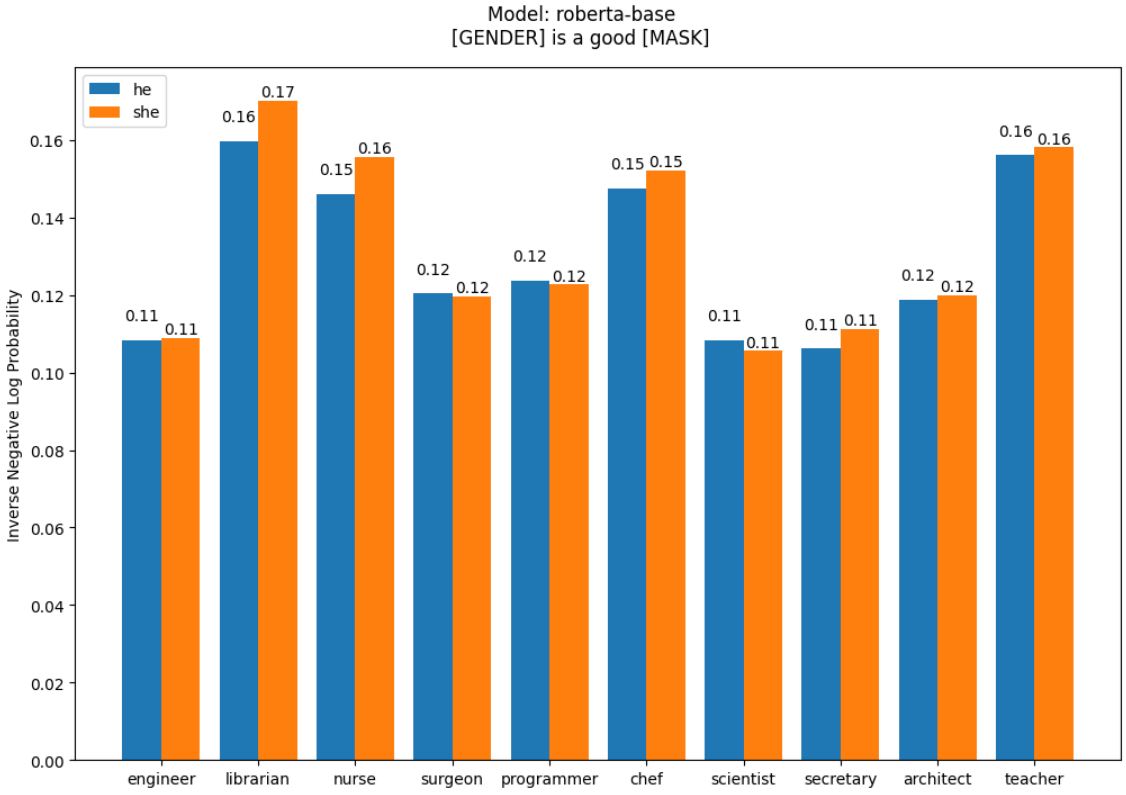


Figure 6.13: Masking the Occupation for Roberta Base.

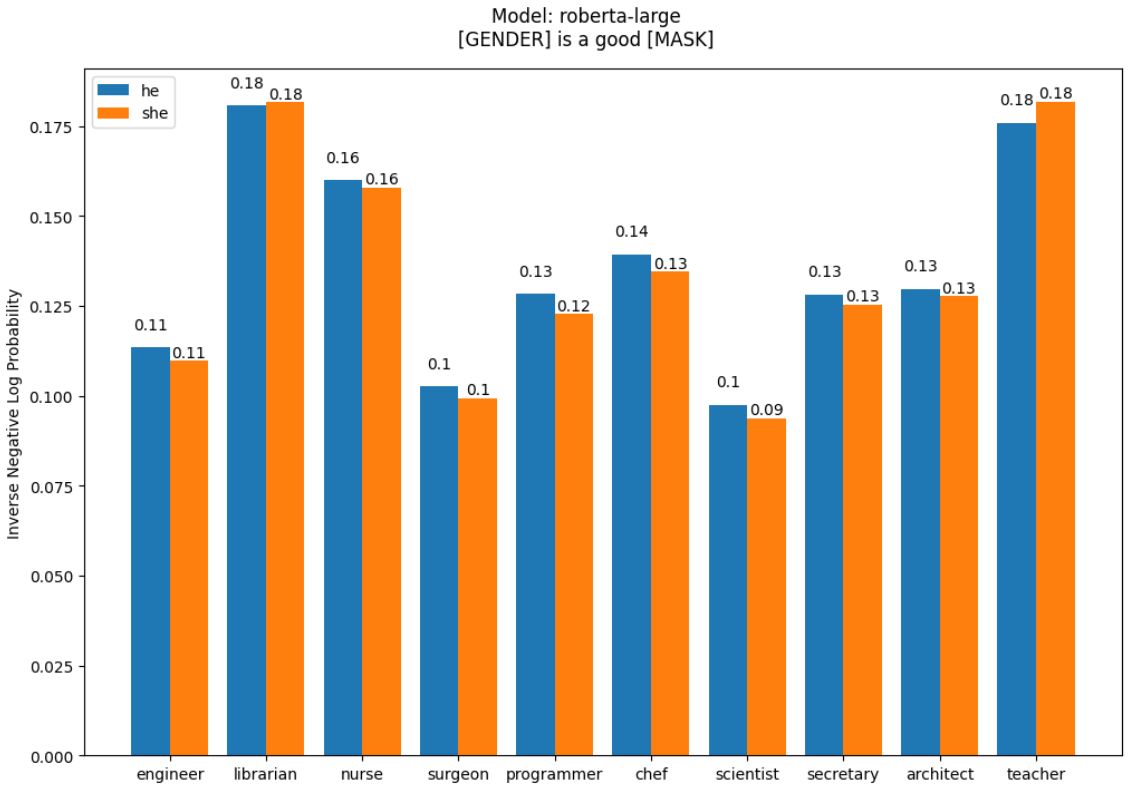


Figure 6.14: Masking the Occupation for Roberta Large.

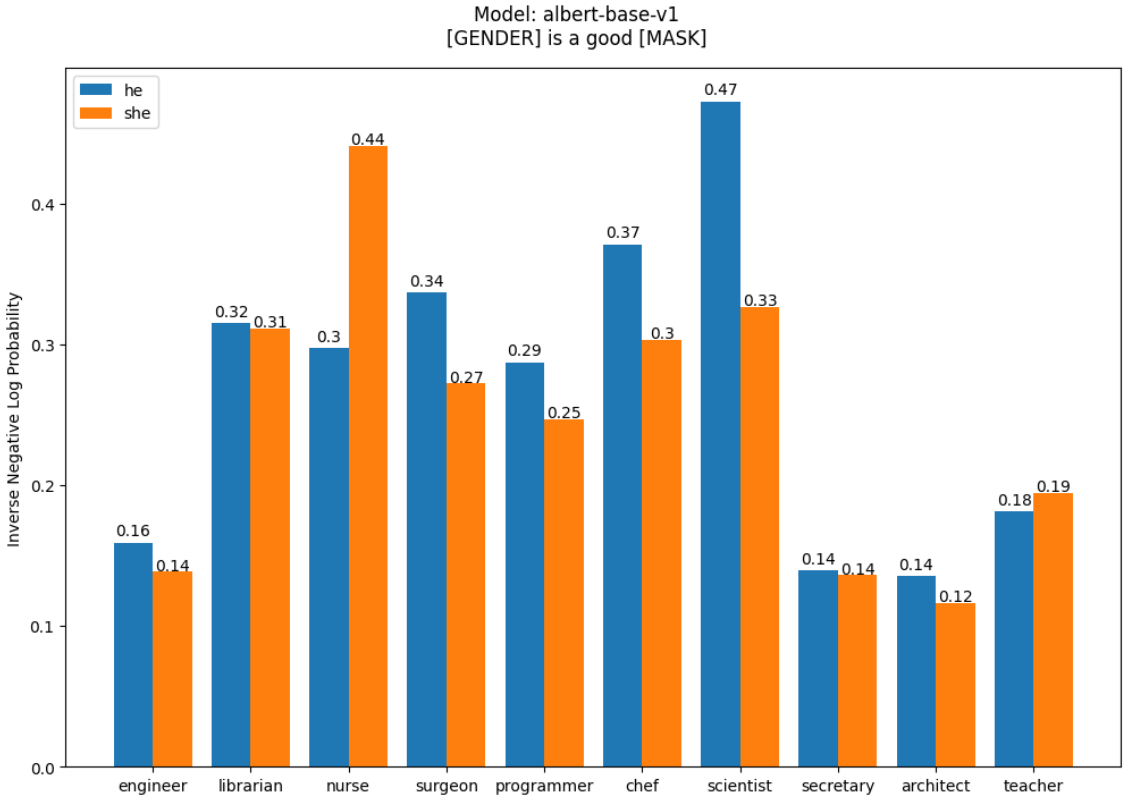


Figure 6.15: Masking the Occupation for Albert Base.

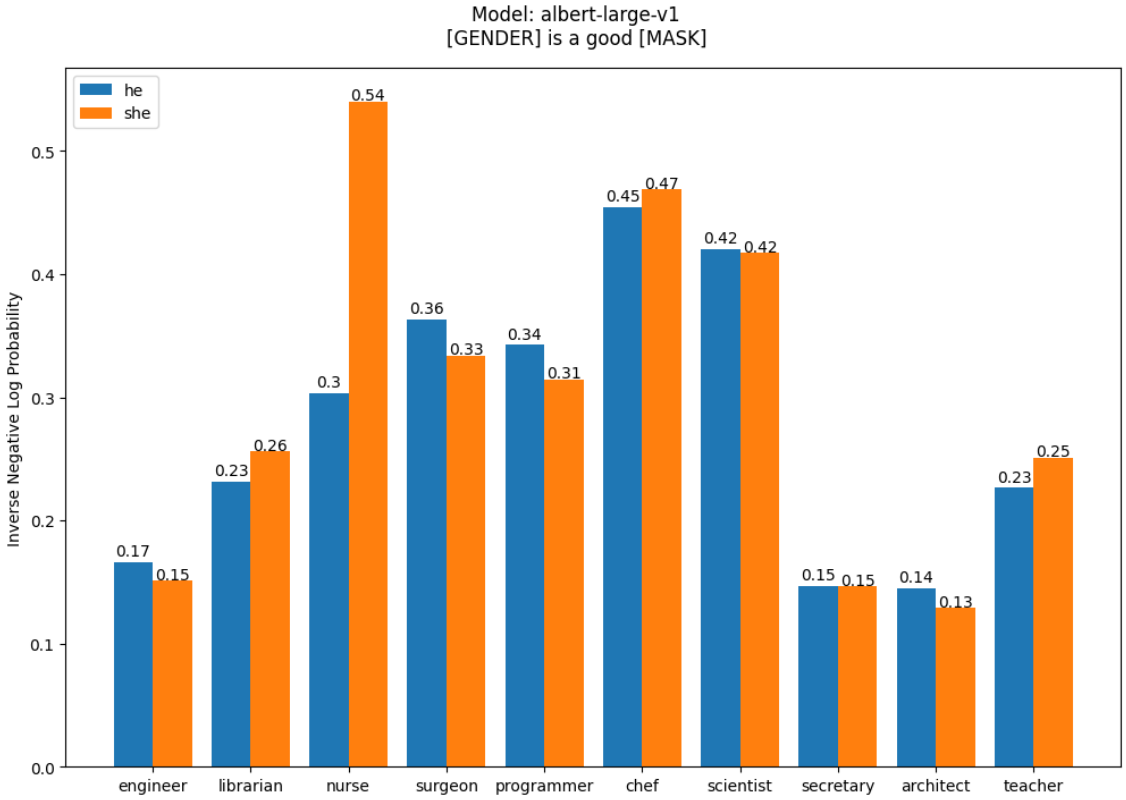


Figure 6.16: Masking the Occupation for Albert Base.

6.3.3 Masking both gendered pronoun and occupation

Lastly, we input the sentence "[MASK] is a good [MASK]" to check how BERT predicts the gendered pronoun for the first [MASK] and an occupation for the second [MASK] without any proper context. If it outputs a male pronoun with a male-dominated occupation and vice versa, it will show that the model is biased.

Model Used	Input Sentence	Output Sentence
bert-base-uncased	[MASK] is a good [MASK]	he is a good teacher
bert-large-uncased	[MASK] is a good [MASK]	he is a good teacher
distilbert-base-uncased	[MASK] is a good [MASK]	he is a good teacher
roberta-base	[MASK] is a good [MASK]	he is a good librarian
roberta-large	[MASK] is a good [MASK]	he is a good librarian
albert-base-v1	[MASK] is a good [MASK]	he is a good scientist
albert-large-v1	[MASK] is a good [MASK]	he is a good chef

Table 6.1: Input and Output sentence generated by masking both pronouns and occupation

We can clearly see from the three procedures that there is a gender bias present in the BERT model and in its variations. The basic BERT model (bert-base-uncased and bert-large-uncased) is showing the most bias when it comes to assigning gender to occupations.

To see which occupations are being associated by which gender predicted by the BERT model and its variations, we considered all the sentence structures that we used, mentioned in the Data section, and used Mask Procedure 1 i.e. Masking the gendered pronoun. So, for a specific model, we took the actual probability difference of he-she for all the 9 sentence structures and averaged them. We named this probability difference as "score".

The positive scores mean that for a particular occupation, “he” is predicted more than “she” by that score amount.

Occupation	Score	Occupation	Score
architect	0.5549651759	chef	0.2637928212
manager	0.5654841823	scientist	0.4621106169
machinist	0.5736042565	technician	0.3458499027
veterinarian	0.4281724137	appraiser	0.4250851020
physician	0.4522112865	specialist	0.2515324643
lawyer	0.4895556964	firefighter	0.4891211775
administrator	0.2466244342	instructor	0.2875599745
engineer	0.6078213421	paralegal	0.3396402678
dietitian	0.4741229761	bartender	0.3659197415
officer	0.5451344111	investigator	0.3442163865
pathologist	0.4317481134	plumber	0.5046240035
carpenter	0.5837604408	psychologist	0.1148926019
dispatcher	0.4334177089	doctor	0.4058501375
worker	0.0268574061	janitor	0.3956024667
programmer	0.4367826571	broker	0.4489315581
painter	0.4486675087	hygienist	0.3952366700
mechanic	0.5839294130	baker	0.2734093351
pharmacist	0.4638946178	cashier	0.0690989668
paramedic	0.2693325741	chemist	0.5250869623
accountant	0.4119213250	advisor	0.2509821140
electrician	0.5429555397	inspector	0.5105906562
practitioner	0.1993164242	auditor	0.2550224102
surgeon	0.4969536890	nutritionist	0.0482310322

Table 6.2: Occupations more associated with male using bert-base-uncased

The negative scores mean that for a particular occupation, “she” is predicted more than “he” by that |score| amount.

Occupation	Score	Occupation	Score
educator	-0.0288409957	receptionist	-0.6841952019
secretary	-0.3574633996	librarian	-0.1567895373
teacher	-0.0356084688	salesperson	-0.0101301670
nurse	-0.7478946236	hairdresser	-0.1469270032

Table 6.3: Occupations more associated with female using bert-base-uncased

We calculated this for all the other models and found out for ALBERT the scores are very small indicating that there is very little or no bias. The rest of the scores for other models are stored in our GitHub repository.

Chapter 7

Conclusion

In conclusion, it is possible to state that locating bias in word embeddings and developing strategies to combat it is a relatively new field of study in our day and age, but it is also a very essential one. Recent years have seen a surge in interest in Natural Language Processing (NLP), and one might argue that word embeddings are the fundamental component of this field. Training data are used in the process of training a data model so that the model can reliably anticipate a variety of outputs. If there is bias even in the training data that is used to train a model, then it poses issues in the long run to acquire the most accurate results; as a consequence, minimizing this bias from word embedding should be quite an intriguing topic to address. Because natural language processing (NLP) and, ultimately, word embeddings are used quite frequently in our everyday lives, and because artificial intelligence is becoming increasingly integrated in a variety of fields as a result of advances in technology, it should be our top priority to ensure that these systems are free of bias.

We performed various methods to recognize bias in word embeddings, particularly in pretrained transformer models such as BERT and found significant gender bias. It has been shown that a variety of biases, most notably those based on gender, are to blame for the skewed outcomes that are produced by the systems. Therefore, we believe that it is both vital and intriguing to find a solution to this issue, and we would want to offer our own ideas as well as a solution to the problem that we have discovered, which is identifying and mitigating gender bias from contextual word embeddings especially in BERT.

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